Ali Tavakoli

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EDUCATION

Sharif University of Technology Tehran, Iran
M.Sc. in Computer Science Expected Fall 2026

- Admitted through the **Exceptional Talents** provision (national quota of only 2 students), which allows top-ranked undergraduates to enter the M.Sc. without taking the entry exam.

University of Tehran Tehran, Iran
B.Sc. in Computer Science, Minor in Mathematics (Expected Graduation: 2026) Sep 2022 – Present

- **GPA:** 19.26/20 | **Rank:** 2nd out of 60 students.
- **Computer Science (selected):** Design & Analysis of Algorithms (20/20), Data Structures & Algorithms (19.5/20), Graph Theory (19.1/20), Combinatorics (20/20), Computability Theory (20/20), Mathematical Logic (19.8/20), Advanced Programming (19/20), Fundamentals of Computer Science and Programming (20/20), Fundamentals of Theory of Computation (19.75/20).
- **Mathematics (selected):** Linear Algebra (20/20), Mathematical Analysis I–II (20/20), Convex Analysis (Graduate-level, 20/20), Linear Programming (20/20), Stochastic Processes (20/20), Game Theory (17.25/20).

HONORS AND AWARDS

- **Top 0.5%** in the National University Entrance Exam (Konkur) among 100,000+ candidates (Math & Engineering).
- **Iranian National Olympiad in Informatics (INOI):** National finalist and recipient of the Diploma of Honor (Top 80 in Iran).

RESEARCH EXPERIENCE

Undergraduate Research in Mathematical Optimization Ongoing
Advisor: Dr. Soleimani Damaneh, University of Tehran

- Surveying and comparing different techniques for solving **integer programming** problems.
- Studying **duality** in integer programming, including Lagrangian and superadditive duality.

INDEPENDENT STUDY

Reading Research Papers in Algorithms, Graph Theory, and Game Theory

- Studied a paper by Li, Peng, Wang, and Zhou on fairness-aware facility location models with externalities.
- Read Kim et al.'s paper on LOCAL certification of MSO2 properties in treewidth-bounded graphs – the part connecting logic to distributed algorithms was the most interesting to me.
- Went through a paper by Bateni, Esfandiari, HosseinGhorban, and Montaseri on density-based active covering methods.

RESEARCH INTERESTS

Theoretical computer science, especially graph theory, algorithms, and combinatorics. Related interests include combinatorial optimization and integer programming.

ACADEMIC SERVICE

- **Grader, Iranian National Olympiad in Informatics (INOI)** Summer 2024

TEACHING EXPERIENCE

University of Tehran

- **Head Teaching Assistant**
  - **Graph Theory** (2 semesters) – Dr. M. Mohammad-Noori
  - **Fundamentals of Combinatorics** – Dr. M. Mohammad-Noori
  - Prepared assignments and quizzes, coordinated grading, and led weekly problem sessions for 40+ students.
- **Teaching Assistant**
  - **Algorithms:** Design & Analysis of Algorithms, Data Structures & Algorithms
  - **Mathematics:** Linear Algebra, Mathematical Analysis, Calculus II
  - **Programming:** Advanced Programming (C++), Basic Programming (C)

Sharif University of Technology (MicroMaster Platform)

- **Teaching Assistant**
  - **Discrete Mathematics** – Dr. Abam
  - Recorded comprehensive video tutorials detailing step-by-step solutions for assigned course exercises on the university's open education platform.

WORKSHOPS & ADVANCED TRAINING

Nesin Mathematics Village August 2026

- **Probability in Turkey 2026:** Selected for this competitive summer school; fully funded, including travel and accommodation.

Institute for Research in Fundamental Sciences (IPM) Summer 2023

- **Summer Mathematics School:** Lectures on analytic number theory and random graphs.

SELECTED PROJECTS

- **Bayesian Inference via MCMC (Metropolis-Hastings)** Sep 2025
  - Implemented the Metropolis-Hastings algorithm for sampling from complex distributions.
  - Used MCMC for parameter estimation in a linear model.
- **SAT-Based Solver for Sokoban** (GitHub) June 2024
  - Modeled the puzzle using propositional logic and reduced it to SAT.
  - Converted constraints to CNF and decoded solution paths using a SAT solver.
- **Multi-Armed Bandit Simulation (Bakery Problem)** (GitHub) July 2023
  - Implemented Thompson Sampling using Beta-Binomial models.
  - Simulated stochastic rewards and evaluated performance in Python (NumPy, SciPy).
- **Competitive Programming:** Active participant on Codeforces and AtCoder. Maximum Codeforces rating 2007 (Candidate Master), Maximum AtCoder rating 1437 (3 Kyu). Experience with graph algorithms, dynamic programming, and number theory. Profiles: Codeforces | AtCoder

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C
- **Tools:** LaTeX, Git, Linux
- **Languages:** Persian (Native), English (TOEFL iBT 102: R29, L28, W24, S21)